Patient Flow Simulator — Decision Report

Type 1 Emergency Department

Executive summary

- Modelled demand: 308 attendances/day (1,851 patients over the simulated period).
- 4-hour standard: 75% [BELOW the >=78% target]; 95% CI 70-80% over 24 runs; 455 breaches.
- Mean time in department: 3.3 h (95% CI 3.0-3.5 h); 95th percentile 8.7 h.
- Admitted: 17% of attendances.

Bottleneck diagnosis

Biggest delay: SDEC — patients wait 119 min on average (763 patient-hours lost over the run).

Recommendation: sdec doctor is busy (79% utilisation) and likely drives this queue; trial extra capacity in Scenario comparison.

UEC Plan 2025/26 scorecard

4-hour standard: 75% (target >=78%)	**AMBER**
12-hour waits: 2% (target <10%)	**GREEN**
Ambulance handover <45 min: 95% (target >=95%)	**GREEN**
Mental health >24h in ED: 54% (target <5%)	**RED**
Children's 4-hour standard: 75% (target >=80%)	**AMBER**

What-if scenarios (ranked)

Scenario	4-hr perf.	vs baseline	Mean LOS
Baseline	74%	—	3.3 h
+3 sdec doctor	84%	**+10 pts**	2.8 h
+8 inpatient bed	75%	**+1 pts**	3.2 h
Winter +15% demand	68%	**-6 pts**	4.0 h

For demonstration — synthetic data seeded from the beta config.json. Validate against local data before operational use.

Patient Flow Simulator

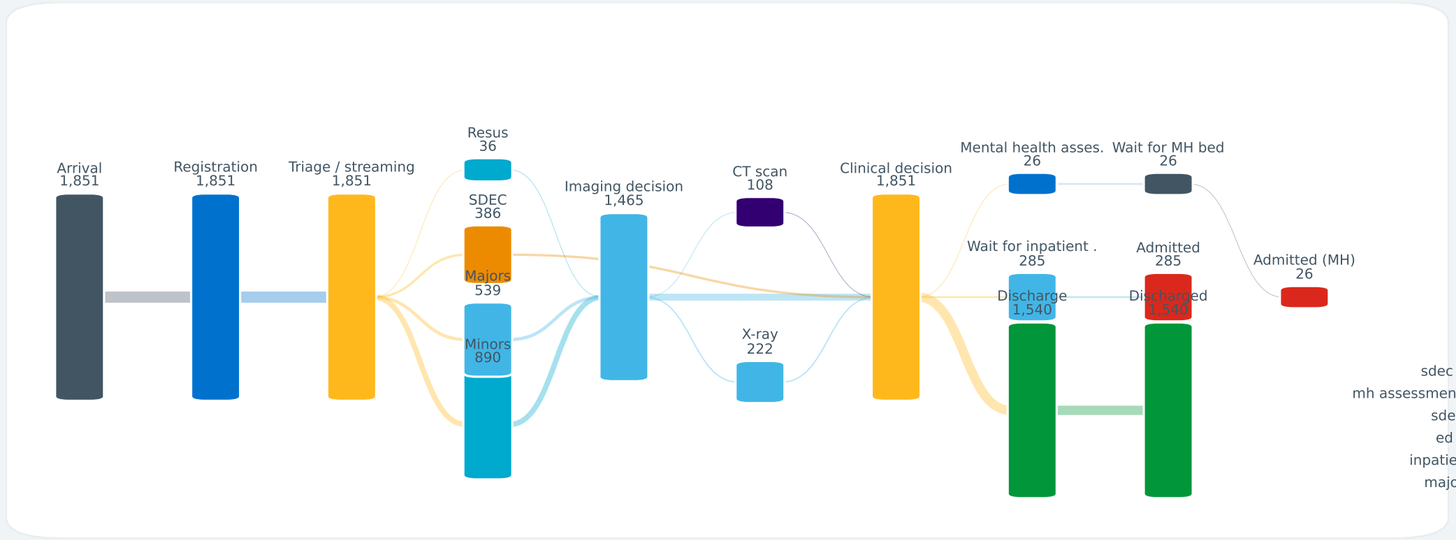
Discrete-event simulation for emergency & urgent care planning

Type 1 Emergency Department

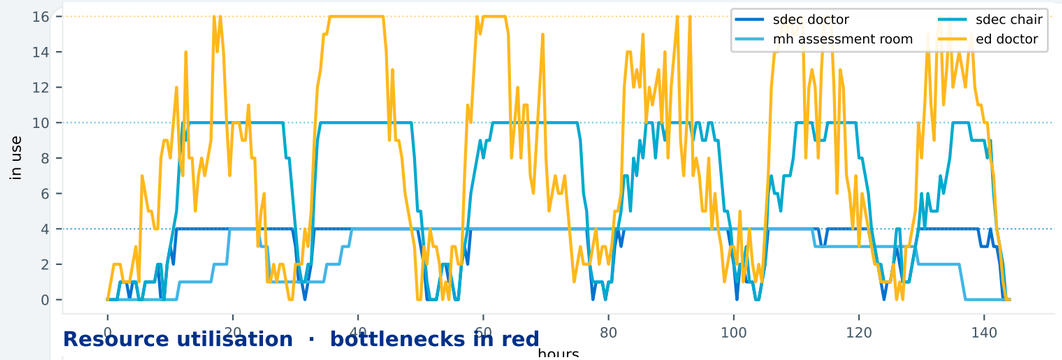
308 attendances/day · 6-day run · 24 Monte-Carlo replications

4-HOUR PERFORMANCE	MEAN TIME IN DEPT	95TH PERCENTILE LOS	ADMITTED	DAILY ATTENDANCES
75%	**3.3h**	**8.7h**	**17%**	**308**
95% CI 70-80%	95% CI 3.0-3.5h	longest typical wait	9.2h mean LOS	1,851 simulated

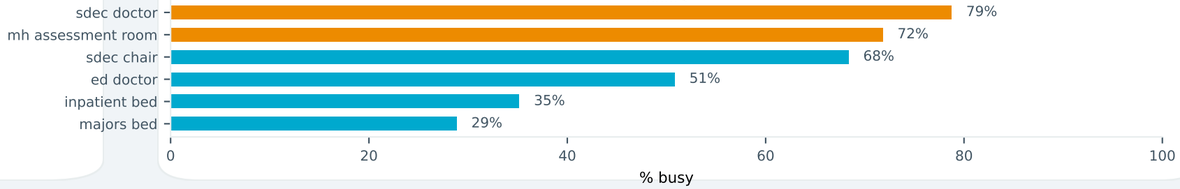
Patient flow · arrival to disposition



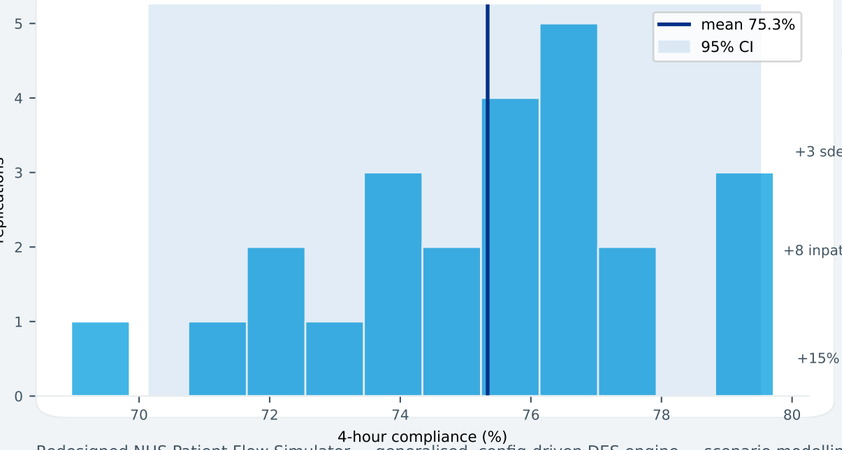
Resource occupancy vs capacity (dotted)



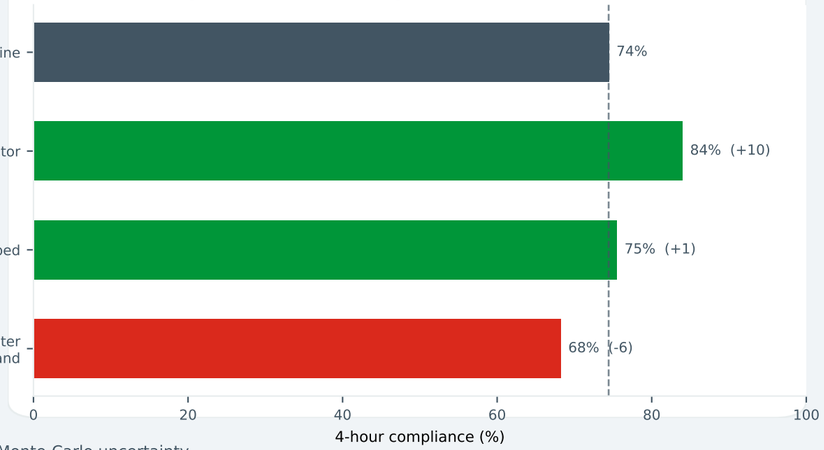
Resource utilisation · bottlenecks in red



Monte-Carlo: 4-hour performance variability



Scenario comparison · 4-hour performance



Where patients wait · mean queue time

STAGE	WAIT	SEEN
Mental health assessment	**711m**	26
SDEC	**119m**	386
CT scan	**7m**	108
Minors	**6m**	890
Majors	**5m**	539
Resus	**5m**	36

For demonstration — synthetic data